

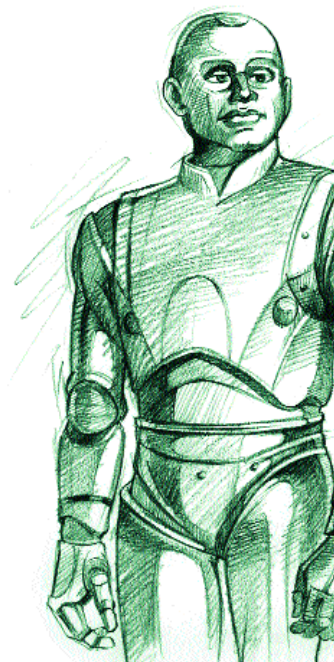
features ■ technology next

We grew up with the *Jetsons* family and half expected the wonderful world of flying cars, automated sidewalks and the floating buildings to come alive. Will robotics bring Rosie the robot maid out of the television screen and into our homes?



Probotics, a small company, has developed Cyé—an answer to domestic woes that took 10 years in the making. Cyé resembles a dustpan on wheels and is designed to serve you dinner and clean the carpet. It uses a personal computer as the hub of its tasks. The user defines the chores that need to be performed, by dragging and dropping icons within a computer program and Cyé takes over from there. Before bending its back though, the robot needs to know the lay of the land. This task requires the owner's input, so that the robot is guided from one extreme of a room to another using a mouse. It then explores between boundaries thus defined,

interpolates the data from its treks and conjures up a map of the space in question. Once mapped, the machine navigates using a pair of wheels that double up as rudimentary sense organs. Whenever the wheels encounter resistance to their motion, the force is translated into electronic signals that the machine can understand. Cyé can therefore detect obstacles in its path using these wheels. Thus aided, the robot will arm itself with a vacuum cleaner and attack the carpet with the vigour reserved for mechanical nuts and bearings. When it runs low on power, it stops what it is doing and heads to the nearest recharging pod for a breather.



Go back to the 1986 hit movie, *Short Circuit*, starring an emotionally wrought, self-aware, existential robot, Number 5. Can AIBO, Sony's replacement dog, ever hope as emotive and expressive as that movie's star?

The inimitable series by Douglas Adams, *The Hitchiker's Guide to the Galaxy*, had a peculiar element to it—an alien fish, which when appropriately inserted could instantaneously translate between any and every language. Termed Babelfish, will the real world ever hold it with its ear?



The German Research Center for Artificial Intelligence (DFKI) along with IBM and Siemens has jointly developed an AI system that aims to bridge the language gap. Language resources such as lexicons, grammar, and test suite form the foundation of the technology. Built upon this are components for the design and implementation of a language such as word 'taggers', sentence parsers and sentence generators, along with tools for the maintenance of terminologies and dictionaries. Finally, AI routines that help in speech analysis and synthesis are brought to task. The entire system performs a bevy of tasks including information extraction from spoken

sentences, cross-lingual information access, linguistic analysis and grammar checking. The system has been commercially implemented in the form of Verbmobil, a cellular phone service that can translate spoken English, German, Japanese and Chinese in a few milliseconds at an accuracy rate of close to 90 per cent.

Better AI routines are all that stop speech synthesizers from becoming universal translators. Spoken words are a part of the communication process, which includes gestures and body language. The ultimate translator would thus be one that analyses not only words but also the slight tilt of the head that separates a comment from a conde-

Kismet is a robot, complete with eyes, eyebrows, ears and an experiment that is destined to be adorable, if not great things. Developed by a Massachusetts Institute of Technology (MIT) team headed by Rodney Brooks, the robot is designed to self-learn just like a two-year-old human toddler. Traditionally, AI has been about implementing so-called algorithms that define a set of rules to replicate the world and a set to define the robot's behaviour within the simulated world. Kismet thinks differently. Rules are defined, but are emergent. Kismet is built on a tree-structure of simple components; the complex behaviour is only visible on a large scale—the robot can communicate its desires using a range of expressions that cover interest, happiness, surprise and anxiety. Such techniques can equip us with the tools to implement complex interaction and feedback systems in robots of tomorrow.

scending remark.